

Vector Fields Generated by Point Source

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1. Abstract

By iteratively applying the Helmholtz decomposition theorem, this paper derives a novel class of vector field models, which may find applications in describing charge dynamics, electromagnetic fields, and celestial motion.

2. Notations and Conventions

We identify $\mathbb{R}^4$ with $\mathbb{R}^3 \times \mathbb{R}$ via the global coordinates (x, y, z, w) , where (x, y, z) denote the standard Cartesian coordinates on $\mathbb{R}^3$ and w is a real parameter. We further adopt the following conventions:

$$\rho \triangleq \sqrt{x^2 + y^2 + z^2},$$

$$\nabla \triangleq \mathbf{i}\partial_x + \mathbf{j}\partial_y + \mathbf{k}\partial_z,$$

where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the unit basis vectors of $\mathbb{R}^3$, and $\partial_x, \partial_y, \partial_z$ represent partial derivatives with respect to the first three coordinate variables.

3. Helmholtz Decomposition Theorem

Let $\mathbf{F}: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be a vector field with component functions $F_x, F_y, F_z \in C^1(\mathbb{R}^4)$. If the decay condition

$$\lim_{\rho \rightarrow +\infty} \rho \|\mathbf{F}\| = 0$$

holds, then there exist a scalar $\alpha \in \mathbb{R}$, a scalar potential $\phi: \mathbb{R}^4 \rightarrow \mathbb{R}$, and a vector potential

$\mathbf{A}: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ such that

$$\mathbf{F} = \nabla\phi - \alpha\nabla \times \mathbf{A}.$$

4. Iterated Helmholtz Decomposition

Suppose $\mathbf{M}_0: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ is differentiable with respect to w , and its w -derivative admits the component decomposition

$$\partial_w \mathbf{M}_0 = \partial_w M_{0x} \mathbf{i} + \partial_w M_{0y} \mathbf{j} + \partial_w M_{0z} \mathbf{k}.$$

Assume further that $\partial_w M_{0x}, \partial_w M_{0y}, \partial_w M_{0z} \in C^1(\mathbb{R}^4)$ and satisfy the decay condition

$$\lim_{\rho \rightarrow +\infty} \rho \|\partial_w \mathbf{M}_0\| = 0.$$

Then there exist $\alpha_1 \in \mathbb{R}$, a scalar potential $\phi_1: \mathbb{R}^4 \rightarrow \mathbb{R}$, and a vector field $\mathbf{M}_1: \mathbb{R}^4 \rightarrow \mathbb{R}^3$ such that

$$\partial_w \mathbf{M}_0 = \nabla \phi_1 - \alpha_1 \nabla \times \mathbf{M}_1.$$

Iterating this construction under the standing assumptions of w -differentiability, C^1 -smoothness, and decay at infinity yields the recursive relation

$$\partial_w \mathbf{M}_n = \nabla \phi_{n+1} - \alpha_n \nabla \times \mathbf{M}_{n+1} \quad (1)$$

for all $n \in \mathbb{N}_0$.

5. First Canonical Form

Assume the initial field is irrotational:

$$\nabla \times \mathbf{M}_0 = \mathbf{0}, \quad (2)$$

and the coupling coefficients are constant across iterations:

$$\alpha_n \triangleq \alpha, \quad n \in \mathbb{N}^*.$$

Define the aggregated scalar potential and vector field as

$$\phi \triangleq \sum_{n=1}^{+\infty} \phi_n, \quad \mathbf{M} \triangleq \sum_{n=0}^{+\infty} \mathbf{M}_n.$$

Summing the recursive relations (1) and invoking (2) yields the closed evolution equation

$$\partial_w \mathbf{M} = \nabla \phi - \alpha \nabla \times \mathbf{M}. \quad (3)$$

6. Second Canonical Form

Retain the irrotational condition (2), and impose a periodic pattern on the coupling coefficients:

$$\alpha_{2n-1} \triangleq \alpha, \quad \alpha_{2n} \triangleq \beta, \quad n \in \mathbb{N}^*.$$

Introduce the aggregated potentials and vector fields:

$$\begin{aligned}\phi &\triangleq \sum_{n=1}^{+\infty} \phi_{2n-1}, & \psi &\triangleq \sum_{n=1}^{+\infty} \phi_{2n}, \\ \mathbf{M} &\triangleq \sum_{n=1}^{+\infty} \mathbf{M}_{2n-2}, & \mathbf{N} &\triangleq \sum_{n=1}^{+\infty} \mathbf{M}_{2n-1}.\end{aligned}$$

Aggregating the appropriate subsequences of (1) yields the coupled system:

$$\begin{aligned}\partial_w \mathbf{M} &= \nabla \phi - \alpha \nabla \times \mathbf{N}, \\ \partial_w \mathbf{N} &= \nabla \psi - \beta \nabla \times \mathbf{M}.\end{aligned}\tag{4}$$

7. Third Canonical Form

Under the irrotational condition (2), assume the coupling coefficients follow a period-3 pattern:

$$\alpha_{3n-2} \triangleq \alpha, \quad \alpha_{3n-1} \triangleq \beta, \quad \alpha_{3n} \triangleq \gamma, \quad n \in \mathbb{N}^*.$$

Define the aggregated quantities:

$$\begin{aligned}\phi &\triangleq \sum_{n=1}^{+\infty} \phi_{3n-2}, & \psi &\triangleq \sum_{n=1}^{+\infty} \phi_{3n-1}, & \xi &\triangleq \sum_{n=1}^{+\infty} \phi_{3n}, \\ \mathbf{M} &\triangleq \sum_{n=1}^{+\infty} \mathbf{M}_{3n-3}, & \mathbf{N} &\triangleq \sum_{n=1}^{+\infty} \mathbf{M}_{3n-2}, & \mathbf{L} &\triangleq \sum_{n=1}^{+\infty} \mathbf{M}_{3n-1}.\end{aligned}$$

The corresponding aggregated system reads:

$$\begin{aligned}\partial_w \mathbf{M} &= \nabla \phi - \alpha \nabla \times \mathbf{N}, \\ \partial_w \mathbf{N} &= \nabla \psi - \beta \nabla \times \mathbf{L}, \\ \partial_w \mathbf{L} &= \nabla \xi - \gamma \nabla \times \mathbf{M}.\end{aligned}\tag{5}$$

8. Higher-Order Canonical Forms

Generalizations to period- k coefficient patterns for arbitrary positive integers k follow analogously.

9. Structural Properties

A. Deterministic Hierarchy

Once $\mathbf{M}_0$ is specified, all subsequent fields $\mathbf{M}_n (n \in \mathbb{N}^*)$ are uniquely determined by (1). The aggregated fields are thus uniquely determined by $\mathbf{M}_0$.

B. Implicit Dependence on Initial Data

The aggregated evolution equations (3), (4), and (5) do not explicitly contain $\mathbf{M}_0$. The initial field influences the solution solely through the implicit determination of the potential terms and higher-order fields.

C. Parameter Independence

Consider the second canonical form (4) under the auxiliary conditions:

$$\nabla \cdot \mathbf{M} = 0, \quad \nabla \phi = \mathbf{0}, \quad \nabla \psi = \mathbf{0}.$$

Applying the vector identity $\nabla \times (\nabla \times \mathbf{f}) = \nabla(\nabla \cdot \mathbf{f}) - \nabla^2 \mathbf{f}$ and commuting the w -derivative with the curl operator yield

$$\partial_w^2 \mathbf{M} - \frac{1}{\eta^2} \nabla^2 \mathbf{M} = \mathbf{0},$$

where the parameter $\eta \triangleq 1/\sqrt{-\alpha\beta}$ is independent of the choice of $\mathbf{M}_0$.

10. Example: Point Source Field

Let the initial field be defined by

$$\mathbf{M}_0 \triangleq \kappa D \frac{\mathbf{r} - \mathbf{r}_o}{|\mathbf{r} - \mathbf{r}_o|^3},$$

where $\mathbf{r} = (x, y, z)$, $\mathbf{r}_o = (x_o(w), y_o(w), z_o(w))$, $\kappa, D \in \mathbb{R}^*$, and $w \in \mathbb{R}^+$. Define the set $S_o \triangleq \{(x_o(w), y_o(w), z_o(w)) | w \in \mathbb{R}^+\}$ and the domain $S \triangleq \mathbb{R}^3 \setminus S_o$.

On S , the recursive relation (1) holds, and the irrotational condition (2) is satisfied identically.

Under the period-2 coefficient assumption, the system (4) holds on S . Notably, the aggregated fields $\mathbf{M}$, $\mathbf{N}$ do not explicitly depend on D or $\mathbf{r}_o$.

11. Identities

On the region S defined in section 10, the following identities hold:

$$\begin{aligned} \partial_w \frac{\mathbf{r} - \mathbf{r}_o}{|\mathbf{r} - \mathbf{r}_o|} &= \nabla \times \left[\frac{d\mathbf{r}_o}{dw} \times \frac{\mathbf{r} - \mathbf{r}_o}{|\mathbf{r} - \mathbf{r}_o|^3} \right], \\ \partial_w \left[\frac{d\mathbf{r}_o}{dw} \times \frac{\mathbf{r} - \mathbf{r}_o}{|\mathbf{r} - \mathbf{r}_o|^3} \right] &= \nabla \times \left[\frac{d\mathbf{r}_o}{dw} \times \left[\frac{d\mathbf{r}_o}{dw} \times \frac{\mathbf{r} - \mathbf{r}_o}{|\mathbf{r} - \mathbf{r}_o|^3} \right] + \frac{\frac{d^2 \mathbf{r}_o}{dw^2}}{|\mathbf{r} - \mathbf{r}_o|} \right]. \end{aligned}$$

12. Conclusion

In view of the structural correspondence between the Newtonian gravitational law and the point-

source field defined above, the proposed vector field framework offers a potential new paradigm for modeling celestial dynamics and gravitational waves. When applied to electrodynamics, the formalism provides a novel description of charge motion.